

Safety Data Sheet(SDS)

1. Identification of the substance/mixture and of the company/undertaking

- 1) Product identifier : BUTYL ACRYLATE BULK
- 2) Relevant identified uses of the substance or mixture and uses advised against
- Relevant identified uses
Adhesives, sealants, Coatings and paints, thinners, paint removers, Others (Surface-active agents, Non-woven fabric, synthetic fiber, leather)
 - Uses advised against
Prohibited from use other than recommended
- 3) Supplier information
- Company name [Manufacture]
Company : LG Chem, Ltd.
Address : 58, Yeosusandan 4-ro, Yeosu-si, Jeollanam-do, Republic of Korea
Emergency number : 080-880-0454

2. HAZARD IDENTIFICATION

- 1) Hazard classification
- Flammable liquids Category 3
 - Acute toxicity(Dermal) Category 4
 - Acute toxicity(Inhalation:Vapours) Category 3
 - Skin corrosion/irritation Category 2
 - Serious eye damage/eye irritation Category 2
 - Skin sensitization Category 1
 - Specific target organ toxicity single exposure Category 3(Respiratory tract irritation)
 - Hazardous to the aquatic environment, long-term (chronic) Chronic 3

2) Allocation label elements

Hazard pictograms



Signal word

- DANGER

Hazard statements

H226 Flammable liquid and vapour
H312 Harmful in contact with skin
H315 Causes skin irritation
H317 May cause an allergic skin reaction
H319 Causes serious eye irritation
H331 Toxic if inhaled
H335 May cause respiratory irritation
H412 Harmful to aquatic life with long lasting effects

Precautionary statements

- Prevention

P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P233 Keep container tightly closed.
P240 Ground and bond container and receiving equipment.
P241 Use facilities such as explosion-proof electricity, ventilation and lighting.
P242 Use nonsparking tools.
P243 Take action to prevent static discharges.
P261 Avoid breathing mist/vapours/spray.
P264 Wash eye, skin thoroughly after handling.
P271 Use only outdoors or in a wellventilated area.
P272 Contaminated work clothing should not be allowed out of the workplace.
P273 Avoid release to the environment.
P280 Wear protective gloves/protective clothing/eye protection/face protection.

- Response

P302+P352 If you get on your skin: Wash with a large amount of water.
P303+P361+P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower].
P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P311 Get medical institutions and doctors' consultation.
P312 If you feel uncomfortable, receive medical institutions and doctors' consultation.
P321 If you eat or inhale the substance, do not take artificial breath in the oral cavity, and use appropriate breathing medical equipment. When contacting with substances, make a treatment such as rinsing the skin and eyes in the water flowing more than 20 minutes.
P332+P313 If skin irritation occurs: Get medical advice/attention.
P333+P313 If skin irritation or a rash occurs: Get medical advice/attention.
P337+P313 If eye irritation persists: Get medical advice/attention.
P362+P364 Take off contaminated clothing and wash it before reuse.
P370+P378 화재 시: 불을 끄기 위해 이산화탄소, 물분무, 알콜포말을(를) 사용하십시오.

- Storage

P403+P233 Store in a wellventilated place. Keep container tightly closed.

P403+P235 Store in a wellventilated place. Keep cool.

P405 Store locked up.

- Disposal

P501 Dispose of contents and containers according to the legislation of the waste

3) Other hazards

- No data available

o Product NFPA Level

Health	Flammability	Reactivity
3	2	2

(⌘ 0 = Stable , 1 = Low , 2 = Medium , 3 = High , 4 = Very High)

3. Composition/Information on ingredients

Components	Common name	CAS No.	PCT(wt%)
n-Butyl Acrylate	n-Butyl Acrylate	141-32-2	100

4. FIRST AID MEASURES

1) Following eye contact

- Seek immediate medical assistance.
- In case of contact with substance, immediately flush skin or eyes with running water for at least 20 minutes.

2) Following skin contact

- For minor skin contact, avoid spreading material on unaffected skin.
- Seek immediate medical assistance.
- For hot product, immediately immerse in or flush the affected area with large amounts of cold water to dissipate heat.
- In case of contact with substance, immediately flush skin or eyes with running water for at least 20 minutes.
- Wash skin with soap and water.
- Remove and isolate contaminated clothing and shoes.
- In case of burns, immediately cool the affected area with cold water for as long as possible and do not remove clothing that is stuck to the skin.

3) Following inhalation

- If exposed to excessive levels of dusts or fumes, remove to fresh air and get medical attention if cough or other symptoms develop.
- Seek immediate medical assistance.

- Keep victim warm and quiet.
- Do not perform artificial respiration by mouth-to-mouth method, but use appropriate respiratory medical equipment.
- Move to fresh air.
- Administer oxygen if breathing is difficult.
- Give artificial respiration if victim is not breathing.

4) Following ingestion

- Seek immediate medical assistance.
- Do not perform artificial respiration by mouth-to-mouth method, but use appropriate respiratory medical equipment.

5) Advice to physician

- Ensure that medical personnel are aware of the material(s) involved and take precautions to protect themselves.
- Effects of contact or inhalation may be delayed.

5. FIRE FIGHTING MEASURES

1) Suitable (and unsuitable) extinguishing media

- Suitable extinguishing media
 - Large fire; Water spray/fog, regular foam
 - Small Fire: Dry sand, dry chemical alcohol-resistant foam, water spray, regular foam, CO2 fire extinguisher
- Unsuitable extinguishing media
 - High-pressure water (risk of fire spread).

2) Special hazards arising from the substance or mixture

- Pyrolytic product
 - During burning, irritating and highly toxic gases may be produced by thermal decomposition or combustion.
 - In case of fire, carbon monoxide and/or carbon dioxide and other harmful substances may be formed.
- Risk of fire and explosion
 - May violently polymerize and result in fire and explosion.
 - Can form explosive mixtures at temperatures at or above the flashpoint.
 - Containers may explode when heated.
 - When heated, vapors may form explosive mixtures with air: indoors, outdoors and sewers explosion hazards.
 - HIGHLY FLAMMABLE: Will be easily ignited by heat, sparks or flames.
 - Runoff may create fire or explosion hazard.
 - Vapor explosion hazard indoors, outdoors or in sewers.
 - Vapors may form explosive mixtures with air.
 - Vapors may travel to source of ignition and flash back.

○ Other

- Fire may produce irritating, corrosive and/or toxic gases.
- May cause toxic effects if inhaled or absorbed through skin.

3) Special protective equipment for firefighters

- Rescuers should put on appropriate protective gear.
- Evacuate area and fight fire from a safe distance.
- Cautions ; Most of liquids are lighter than water.
- Most vapors are heavier than air. They will spread along ground and collect in low or confined areas (sewers, basements, tanks).
- Dike fire-control water for later disposal; do not scatter the material.
- Move containers from fire area if you can do it without risk.
- Fire involving Tanks: Fight fire from maximum distance or use unmanned hose holders or monitor nozzles.
- Fire involving Tanks: Cool containers with flooding quantities of water until well after fire is out.
- Fire involving Tanks: Withdraw immediately in case of rising sound from venting safety devices or discoloration of tank.
- Fire involving Tanks: ALWAYS stay away from tanks engulfed in fire.
- Fire involving Tanks: For massive fire, use unmanned hose holders or monitor nozzles; if this is impossible, withdraw from area and let fire burn.

6. ACCIDENTAL RELEASE MEASURES

1) Health considerations and protective equipment

- Do not touch or go near exposed material.
- The very fine particles can cause a fire or explosion, eliminate all ignition sources.
- ELIMINATE all ignition sources (no smoking, flares, sparks or flames in immediate area).
- All equipment used when handling the product must be grounded.
- Prevent dust cloud.
- Clean up spills immediately, observing precautions in Protective Equipment section.
- Stop leak if you can do it without risk.
- Do not touch damaged containers or spilled material unless wearing appropriate protective clothing.
- A vapor suppressing foam may be used to reduce vapors.
- Cover with plastic sheet to prevent spreading.
- Please note that materials and conditions to be avoided.

2) Environmental precautions

- Runoff may cause pollution.
- Prevent entry into waterways, sewers, basements or confined areas.
- Keep out of waterways.

3) For cleaning up

- Absorb or cover with dry earth, sand or other non-combustible material and transfer to containers.
- Reduce airborne dust and prevent scattering by moistening with water.
- Large Spill: Dike far ahead of liquid spill for later disposal.

- Cover powder spill with plastic sheet or tarp to minimize spreading and keep powder dry.
- Absorb spill with inert material (e.g., dry sand or earth), then place in a chemical waste container.
- Small Spill: Absorb with earth, sand or other non-combustible material and transfer to containers for later disposal.
- Dike and collect water used to fight fire.
- Absorb the liquid and scrub the area with detergent and water.
- Use clean non-sparking tools to collect absorbed material.
- With clean shovel place material into clean, dry container and cover loosely; move containers from spill area.

7. HANDLING AND STORAGE

1) Precautions for safe handling

- Avoid breathing vapors from heated material.
- Loosen closure cautiously before opening.
- CAUTION: High temperature.
- Handling refer to engineering control/personal protection section.
- All equipment used when handling the product must be grounded.
- DO NOT PRESSURIZE, CUT, WELD, BRAZE, SOLDER, DRILL, GRIND, OR EXPOSE SUCH CONTAINERS TO HEAT, FLAME, SPARKS, STATIC ELECTRICITY, OR OTHER SOURCES OF IGNITION.
- Caution: Heat.
- Follow all SDS/label precautions even after container is emptied because they may retain product residues.
- Avoid prolonged or repeated contact with skin.
- Measure atmospheric oxygen concentration and ventilate the area during the operation since low-closed area can cause oxygen deficiency.
- Do not enter storage area unless adequately ventilated.
- Use care in handling/storage.
- Please note that materials and conditions to be avoided.
- Use only in a well-ventilated area.

2) Conditions for safe storage (including any incompatibilities)

- Empty drums should be completely drained, properly bunged, and promptly returned to a drum reconditioner, or properly disposed of.

8. EXPOSURE CONTROLS AND PERSONAL PROTECTION

1) Chemical exposure limits, Biological exposure standard

Components	Occupational exposure limits	ACGIH	Biological standard
n-Butyl Acrylate	TWA : 2.0ppm STEL : 10.0ppm	TWA : 2.0ppm STEL : Not applicable	해당없음

2) Appropriate engineering controls

- Use process enclosures, local exhaust ventilation, or other engineering controls to control airborne levels below recommended exposure limits.
- If user operations generate dust, fume, or mist, use ventilation to keep exposure to airborne contaminants below the exposure limit.

- Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.

3) Personal protection equipment

- Respiratory protection
 - Wear breathing protection, which needs a confirmation from the Korea Occupational Safety and Health Agency.
 - In case of insufficient oxygen (<19.6%), wear a supplied air mask or self-contained respirator.
- Eye protection
 - Wear suitable protective goggles and face shields.
- Hand protection
 - Wear suitable protective gloves.
- Body protection
 - Wear suitable protective clothing.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	Clear Liquid
Physical state	Liquid
Colour	Colorless
Odour	Fruity scent
Odour threshold	Fruity scent
pH	No data available
Melting point/freezing point	-64.6 °C
Initial boiling point and boiling range	145°C
Flash point	36°C
Evaporation rate	5.45 mmHg(25°C)
Flammability(solid, gas)	No data available
Upper/lower flammability or explosive limits	9.9 % / 1.3 %
Vapour pressure	5.45 mmHg(25°C)
Solubility(ies)	No data available
Vapour density	No data available
Relative density	No data available
n-octanol/water partition coefficient	2.36
Auto ignition temperature	267 °C
Decomposition temperature	No data available
Viscosity	0.8mm ² /s

Molecular weight(mass)	128.17 g/mol
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10. STABILITY AND REACTIVITY

1) Stability and hazardous reactivity

- May violently polymerize and result in fire and explosion.
- Can form explosive mixtures at temperatures at or above the flashpoint.
- Containers may explode when heated.
- When heated, vapors may form explosive mixtures with air: indoors, outdoors and sewers explosion hazards.
- HIGHLY FLAMMABLE: Will be easily ignited by heat, sparks or flames.
- Runoff may create fire or explosion hazard.
- Vapor explosion hazard indoors, outdoors or in sewers.
- Vapors may form explosive mixtures with air.
- Vapors may travel to source of ignition and flash back.
- Fire may produce irritating, corrosive and/or toxic gases.
- May cause toxic effects if inhaled or absorbed through skin.

2) Conditions to avoid

- Ignition source(heat, spark, flame, etc.).
- Heat, contamination.

3) Incompatible materials

- Combustibles, reducing material.

4) Hazardous decomposition products

- During burning, irritating and highly toxic gases may be produced by thermal decomposition or combustion.
- Corrosive/toxic fume.
- Irritating, corrosive and/or toxic gas.

11. TOXICOLOGICAL INFORMATION

1) Exposure route information

- Inhalation
 - May cause respiratory irritation
 - Toxic if inhaled
- Skin Contact
 - May cause an allergic skin reaction
 - Causes skin irritation
 - Harmful in contact with skin
- Eye Contact
 - Causes serious eye irritation

- Ingestion
 - Not applicable

2) Health hazard information

- Acute toxicity
 - Acute toxicity(Oral)
LD50 3150 mg/kg Test species: Rat (calculated value, OECD TG 401), Source: ECHA
 - Acute toxicity(Dermal)
Id50 >2000 mg/kg Test species: Rabbit, Source: Chemical Substance Information Processing System
 - Acute toxicity(Inhalation:Gases)
No data available
 - Acute toxicity(Inhalation:Vapours)
LC50=10.3mg/L, Sprague-Dawley Rat, OECD TG 403
 - Acute toxicity(Inhalation:Dust/mist)
No data available
- Skin corrosion/irritation
Irritating, Vienna White rabbit
- Serious eye damage/eye irritation
Irritation moderate-severe, Rabbit
- Respiratory sensitization
Causes skin sensitization, Mouse, OECD TG 429
- Skin sensitization
Skin sensitization test using mice: Hypersensitivity (OECD TG 429, EPA OPPTS 870.2600, GLP). Hypersensitivity in guinea pig test. Hypersensitivity in mouse LLNA. Positive in human patch test., Source: ECHA, NITE
- Carcinogenicity
3 (IARC)
A4 (ACGHI)
- Germ cell mutagenicity
In vitro gene mutation test using mammalian cultured cells, negative regardless of metabolic activation (OECD TG 476, GLP), reverse mutation test using microorganisms, negative regardless of metabolic activation (EU Method B.13/14, GLP), chromosomal aberration test using mammalian cultured cells, negative regardless of metabolic activation (OECD TG 473, GLP), no in vivo data available, Source: ECHA
- Reproductive toxicity
Inhalation reproductive toxicity test using rats showed no effect on reproductive organs NOAEC(P)=2.

86 mg/L air Inhalation fetal developmental toxicity test using rats showed changes in maternal body weight, significant decrease in fetal weight, and sporadic malformations occurred slightly at 300 ppm. The incidence of skeletal deformation was similar between the control and treatment groups LOAEC_{maternal} toxicity=ca. 0.52 mg/L air, NOAEC_{fetotoxicity}=ca. 0.52 mg/L air, NOAEC_{teratogenicity}=ca. 1.57 mg/L air (OECD TG 414) In an inhalation developmental toxicity test using rats, resorptions and fetal effects such as a decrease in the number of live fetuses were caused at maternally toxic concentrations. Due to maternal toxicity, NOAEL (maternal) = 25 ppm (0.13 mg/L/day) for weight loss and eye and nose irritation, NOAEL (developmental) = 25 ppm (0.13 mg/L/day) for postimplantation loss, NOAEL (teratogenicity) = 250 ppm. In a separate study, LOAEL maternal = 100 ppm due to decreased absolute weight gain. NOAEL (developmental) = 100 ppm, NOAEL (teratogenicity) = 300 ppm (highest dose tested) due to decreased fetal weight and sporadic teratogenicity in the 300 ppm control group., Source: ECHA, OECD SIDS

○ Specific target organ toxicity single exposure

Short-term exposure effects: Respiratory irritation. If the liquid is swallowed, it may be aspirated into the lungs and cause chemical pneumonitis. Symptoms: Upper respiratory irritation, dyspnea. In an acute inhalation toxicity test on rats exposed for 4 hours, sudden breathing and prone position were observed at 3.6 mg/L. Additionally, at 12.1-16.0 mg/L, clinical symptoms included dyspnea and trembling and closing of the eyelids. In the second inhalation test (LC50=8.08 mg/L), the lungs were congested, and the survivors showed pneumonia. In the third inhalation test (LC50=13.3 mg/L), clinical symptoms including anxiety, dyspnea, congestion of the nasal and ocular mucosa, pulmonary hemorrhage, pulmonary edema, and emphysema were observed., Source: NIOSH IPCS NITE

○ Specific target organ toxicity repeated exposure

As a result of a subchronic repeated oral toxicity test using rats, no deaths or clinical symptoms were observed. NOAEL (male) = 84 mg/kg bw/day, NOAEL (female) = 111 mg/kg bw/day (OECD TG 408, GLP). As a result of a subchronic repeated inhalation toxicity test using rats, body weight decrease was observed as overall toxicity, and clinical chemistry changes were observed only in females. NOAEC (systemic effects) = 0.57 mg/L air, LOAEC (systemic effects) = 1.11 mg/L air, and histological changes were observed in the nasal mucosa as local toxicity. NOAEC (local effects) = 0.11 mg/L air, LOAEC (local effects) = 0.57 mg/L air (OECD TG 413)., Source: ECHA

○ Aspiration hazard

No data available

12. ECOLOGICAL INFORMATION

1) Aquatic toxicity

- Fish

LC50 2.1 mg/l 96 hr Other (Cyprinodon variegatus, flow-through, OECD Guideline 203, GLP), Source: EHCA

- Crustacea

EC50 8.2 mg/l 48 hr Daphnia magna (flow-through, EPA OTS 797.1300, GLP), Source: ECHA

- Aquatic algae

EC50 2.65 mg/l 96 hr Selenastrum capricornutum (exponential, OECD TG 201, GLP), Source: EHCA

2) Persistence and degradation

- n-octanol water partition coefficient

2.36 log K_{ow}, Source: HSDB

- Degradation

No data available

- Biodegradation

80 ~ 90 % 28 day (OECD TG 310, GLP), Source: ECHA

3) Bioaccumulative potential

17.3 (predicted value, calculated using BCFBAF v3.01), Source: ECHA

4) Mobility in soil

No data available

5) Other adverse effects

Crustaceans: *Daphnia magna*, NOEC, 21d, =0.457 mg/L semi-static, OECD TG 211, GLP; Algae: *Selenastrum capricornutum*, NOEC, 96h, <1.8 mg/L exponential, OECD TG 201, GLP, Source: ECHA

13. DISPOSAL CONSIDERATIONS

1) Disposal methods

- Recyclables must be collected in portable bins and disposed of legally according to national regulations.

2) Precautions (including disposal of contaminated container of package)

- Wear an appropriate Personal protection. (See Exposure Controls/Personal Protection section.)
- Avoid direct spillage to rivers, lakes, soil and drains.
- Discuss it according to waste regulation.
- Use a certified waste disposal company.

14. TRANSPORT INFORMATION

1) UN No. : 2348

2) Proper shipping name : BUTYL ACRYLATES, STABILIZED

3) Class or division : 3

4) Packing group : III

5) Marine pollutant : Not applicable

6) Special safety response for transportation or transportation measure :

Emergency measures in case of fire : F-E

Emergency measures in the effluent : S-D

- ADR

· Tunnel restriction code : D/E

- IMDG

- Marine pollutant : Not applicable
- Air transport(IATA)
 - UN No. : 2348
 - Proper shipping name : BUTYL ACRYLATES, STABILIZED
 - Class or division : 3
 - Packing group : III

15. REGULATORY INFORMATION

1) Occupational Safety and Health Act in Korea

Hazardous Substances(Substances Subject to Process Safety Reporting), Substance exposure limits

2) Toxic Chemical Control Act in Korea

- Not applicable

3) Safety Control of Dangerous Substances Act in Korea

Class 4> Second petroleum (water-insoluble)(Designated quantity:1000ℓ)

4) Wastes Control Act in Korea

Not applicable

-In case of disposal, it must be disposed of in accordance with Article 13 of the Waste Management Act.

5) Other regulations in KOREA and Abroad regulations

- ETC regulation
 - Not applicable
 - Not applicable
- Act on the registration and evaluation of chemicals
Existing Commercial Chemical Substances

16. OTHER INFORMATION

1) Reference

2) Date of first issue : 2022-06-07

3) Revision date

- Revised date count : 17
- Last revised date : 2026-01-14

○ Last revised history : BA MSDS 한국 롤셋 영문버전 개정

4) Other